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Experiment on Chinese Remainder Theorem (CRT)

Aim : To implement CRT to perform various operation on large numbers in congruent (Modulo M) using k number of congruences (Modulo m_i)

Understanding :

Chinese Remainder theorem is used for efficient computation in cryptography. In this experiment, I firstly calculated

m_1, m_2, m_3 which denotes the product of all moduli ($M = m_1 \times m_2 \times m_3$). Then, I calculated $M_1 = \frac{M}{m_1}$, $M_2 = \frac{M}{m_2}$, etc. After this, I

calculated the modular inverse of M_1, M_2, M_3 with modulus m_1, m_2, m_3 respectively, which were y_1, y_2, y_3 . The solution thus was calculated. For computing products of two large numbers or any operation between two large numbers, this theorem is widely used. In signals (traffic signals), where we need to find when all signals turn green, CRT algorithm is used which showcases one of the real world examples.

Programming language :- C++.